4. Given an m x n grid and a ball at a starting cell, find the number of ways to move the ball out of the grid boundary in exactly N steps.

Example:

- Input: m=2,n=2,N=2,i=0,j=0 · Output: 6
- Input: m=1,n=3,N=3,i=0,j=1 · Output: 12



5. You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed. All houses at this place are arranged in a circle. That means the first house is the neighbor of the last one. Meanwhile, adjacent houses have security systems connected, and it will automatically contact the police if two adjacent houses were broken into on the same night.

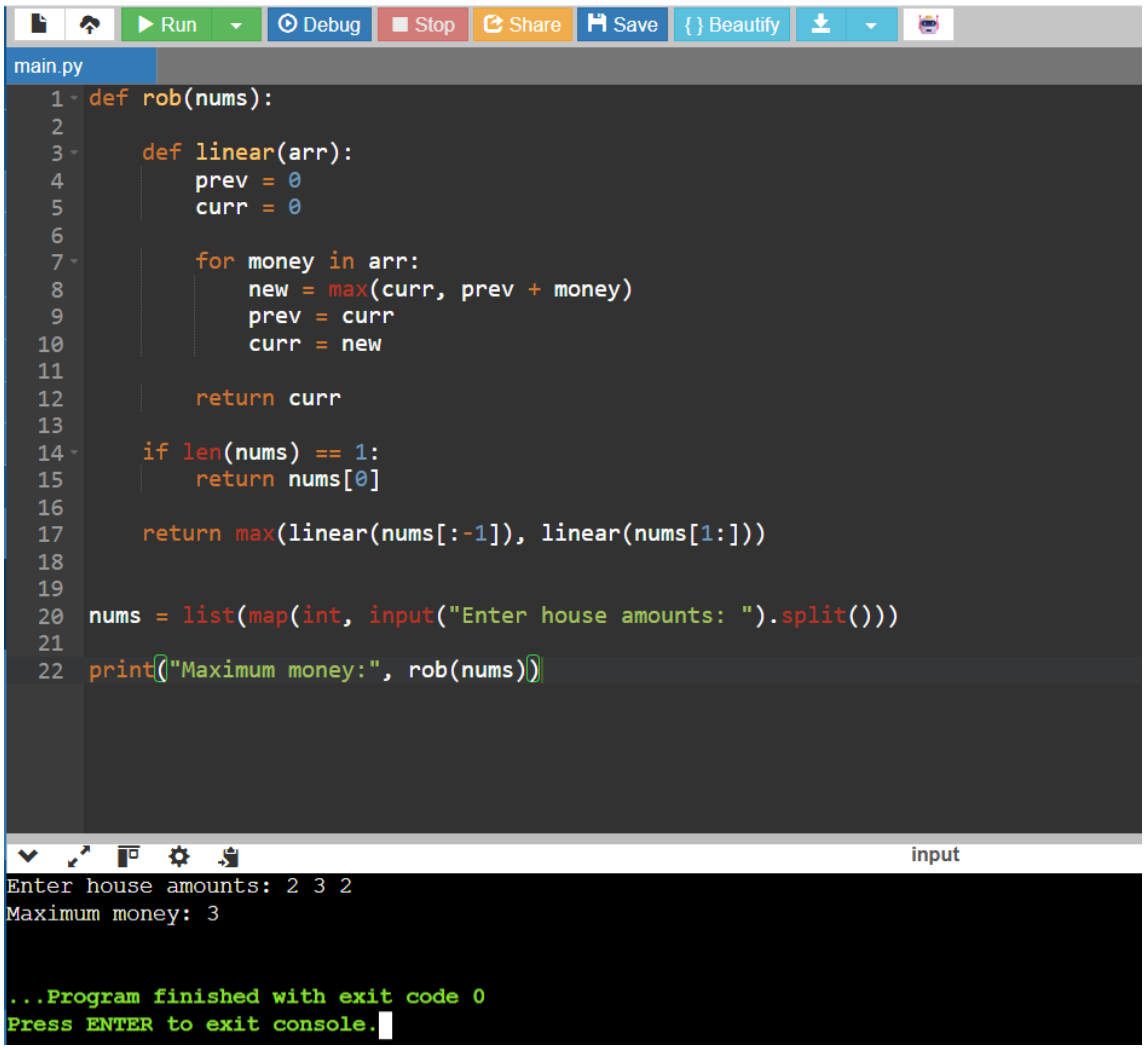
Examples:

Input : nums = [2, 3, 2]

Output : The maximum money you can rob without alerting the police is 3(robbing house 1).

(ii) Input : nums = [1, 2, 3, 1]

Output : The maximum money you can rob without alerting the police is 4 (robbing house 1 and house 3).

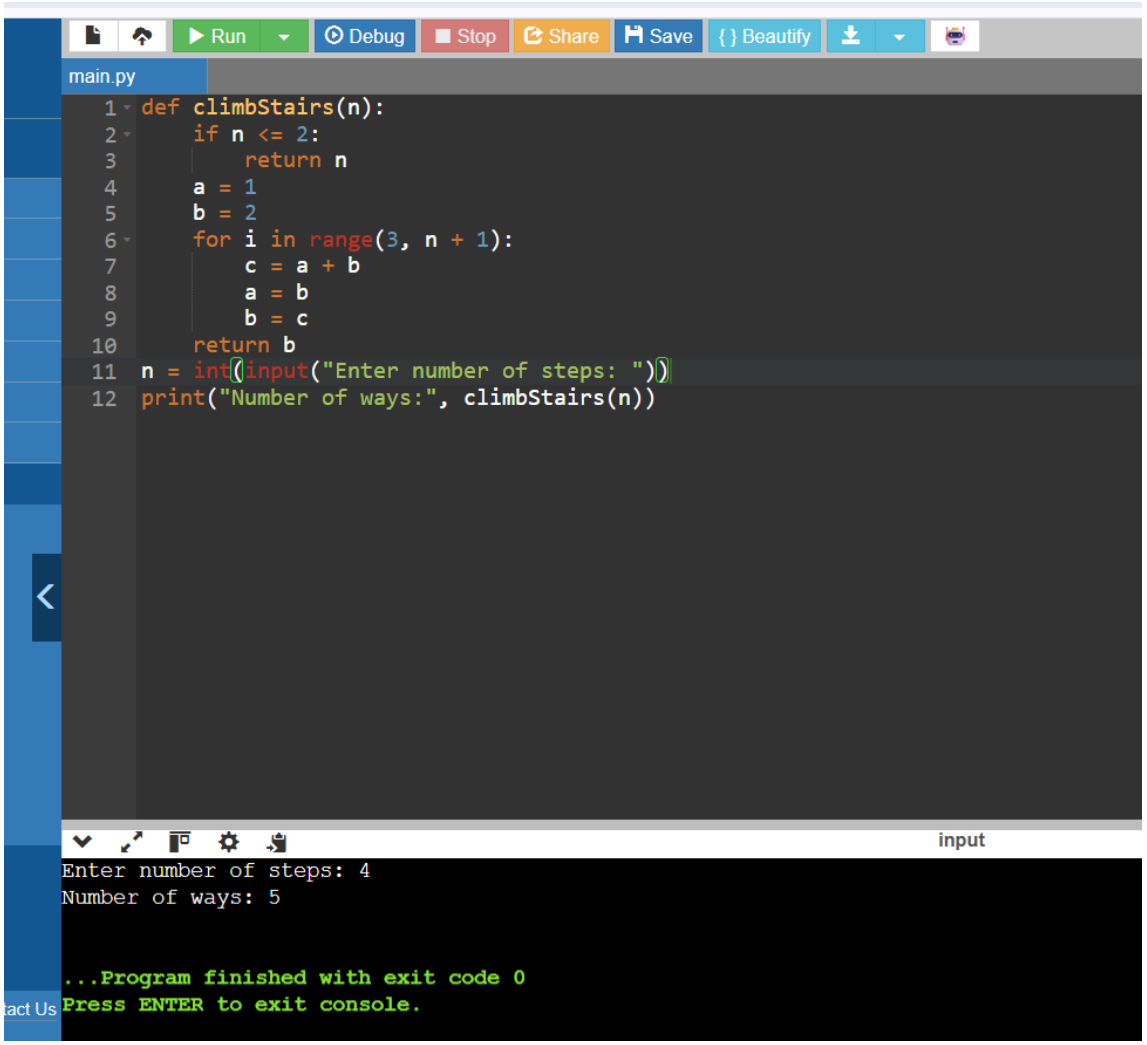


6. You are climbing a staircase. It takes n steps to reach the top. Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?

Examples:

Input: n=4 Output: 5

Input: n=3 Output: 3

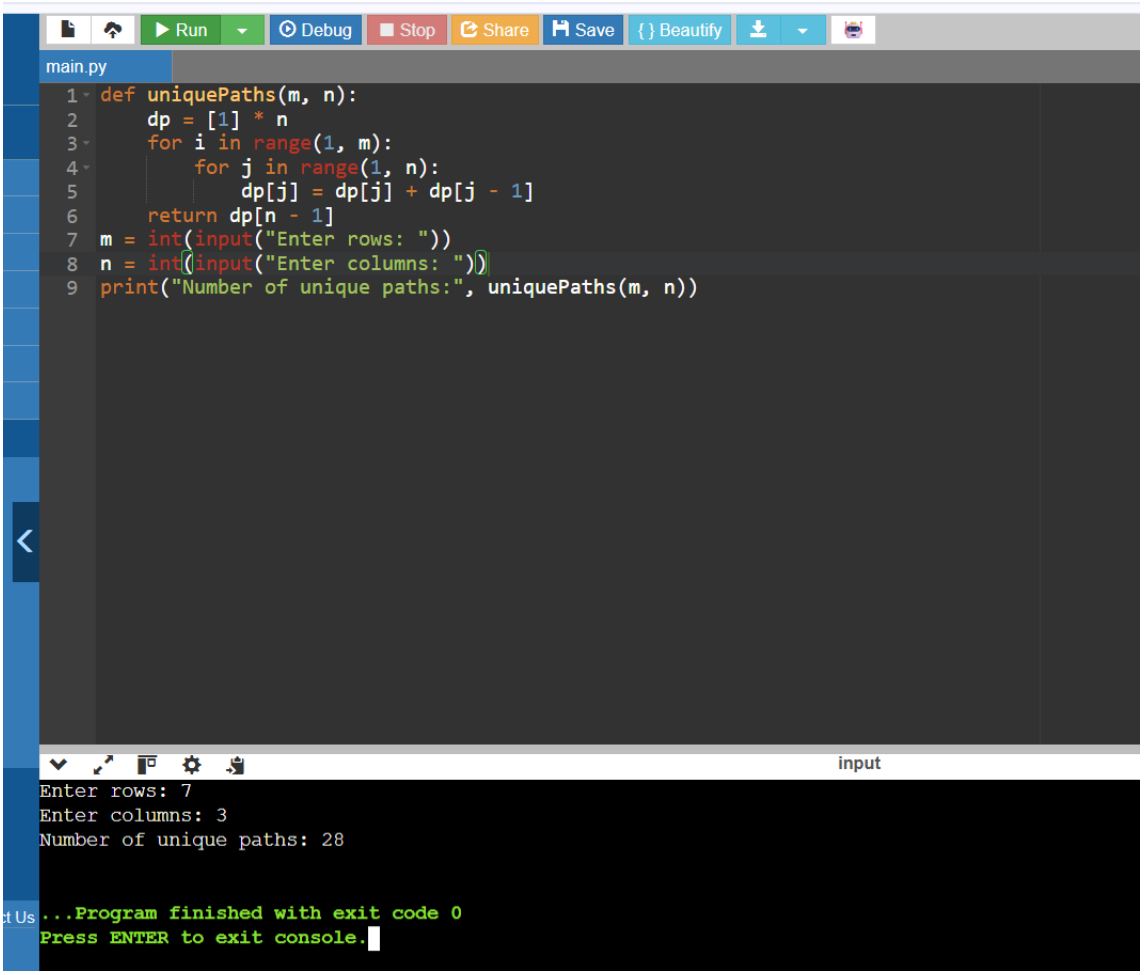


7. A robot is located at the top-left corner of a m×n grid .The robot can only move either down or right at any point in time. The robot is trying to reach the bottom-right corner of the grid. How many possible unique paths are there?

   Examples:

   Input: m=7,n=3 Output: 28

   Input: m=3,n=2 Output: 3


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main.py
def uniquePaths(m, n):
dp = [1] * n
for i in range(1, m):
for j in range(1, n):
dp[j] = dp[j] + dp[j - 1]
return dp[n - 1]
m = int(input("Enter rows: "))
n = int(input("Enter columns: "))
print("Number of unique paths:", uniquePaths(m, n))
input
Enter rows: 7
Enter columns: 3
Number of unique paths: 28
...Program finished with exit code 0
Press ENTER to exit console.